

CCF Database: A Machine-Learning-Annotated Corpus of 266,271 Canadian Climate Articles (1978–2024)

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Abstract

News media shape how societies perceive and respond to climate change, yet large-scale, fine-grained analysis of climate discourse remains hindered by fragmented corpora and ad hoc coding schemes. The Canadian Climate Framing (CCF) database addresses these limitations with a large-scale bilingual annotated corpus and a generalizable annotation framework. It covers 266,271 climate-related articles from 20 Canadian newspapers (1978–2024), processed into 9,198,958 two-sentence units (82.9% English, 17.1% French). Each unit carries 65 hierarchical binary annotations spanning eight main frames, actors, events, solutions, tone, geographic focus, and urgency, produced by 128 BERT and CamemBERT classifiers trained on 4,000+ expert-coded sentences (macro $F_1 = 0.866$, Gwet’s AC1 = 0.894 on a 1,000-sentence gold standard double-coded by an independent annotator). Each category receives an A/B/C reliability tier summarising annotation quality from classifier performance and inter-coder agreement. The database—sentence-level annotations, per-article aggregations (frame proportions, named entities), and 9,462,845 BAAI/bge-m3 sentence embeddings in six relational tables—is openly released on Zenodo (CC-BY 4.0) as a PostgreSQL archive (DOI [10.5281/zenodo.20667151](https://doi.org/10.5281/zenodo.20667151)) and an Apache Parquet bundle (DOI [10.5281/zenodo.20667154](https://doi.org/10.5281/zenodo.20667154)).

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Background & Summary

On February 18, 1978, *The Globe and Mail* ran a story headlined “Some think it’s warming up” that reported on a survey of leading climate experts. Interestingly, almost half a century ago, the article already laid out for its readers what have since become the well-established root causes of climate change:

“The concern is that possible warming, the so-called ‘greenhouse effect,’ is generally associated with the increase of carbon dioxide in the atmosphere by the burning of fossil fuels.”

This excerpt clearly embodies the role the media play in informing society and citizens, who often lack direct access to academic research. At the same time, the media also reflect the social and political realities of the country they cover. Indeed, while some newspapers choose to relay the voices of scientists, others choose to report the controversies, so that the media both inform and reflect political and social realities. Research shows that media shape not only what people think about but also how they think about it, with effects on both public opinion and political agendas [1, 2]. The media are accordingly not a neutral arena for public debate around climate change. Instead, they actively participate in defining and constructing the problems under discussion [3, 4]. Moreover, while digital and social media have grown in prominence, newspapers remain a foundational source of information that continues to shape public discourse and to influence other media formats [5, 6]. Studying print media therefore provides a vantage point on a traditional and still highly influential channel of climate communication. Most importantly, it allows for long-term historical comparison, including periods that predate the rise of social media.

The excerpt above was drawn from the oldest article of the *Canadian Climate Framing (CCF)* database, which we introduce in this Data Descriptor. The CCF database is a machine-learning-preprocessed corpus of 266,271 climate-related newspaper articles published between 1978 and 2024 across 20 major Canadian outlets, accompanied by a rigorous, theoretically grounded annotation framework that we release alongside the data. The corpus contains 9,198,958 two-sentence analytical units, all annotated within a 65-category hierarchical coding scheme. Because Canadian newspaper articles are protected by copyright, the deposit does not redistribute raw sentence text. What is shared instead is (i) the 65-category hierarchical annotation framework—a theoretically grounded coding scheme that integrates main frames, actors, events, solutions, emotional tone, geographic focus, and urgency into a single sentence-level taxonomy—and (ii) the full output of an intensive annotation pipeline built around that framework: 65 binary sentence-level labels per analytical unit, the named-entity extractions (persons, organizations, locations), the article-level aggregates and full metadata (newspaper, date, title, author, page number), the per-category reliability tiers, and the BAAI/bge-m3 sentence embeddings (9.46 million 1024-dimensional vectors). The breadth of the annotation scheme and the embedding layer together make the deposit fully analysable at the sentence level without requiring access to the raw text. By focusing on Canadian print media, this database constitutes a historically influential source of climate information.

Canada offers a particularly compelling case for studying climate discourse. Its federal structure, bilingual culture, and vast territory create a microcosm of the diversity—and

contradictions—found globally. While Canada is highly vulnerable to the impacts of climate change [7], it remains significantly dependent on fossil-fuel extraction and other resource-based industries [8]. This creates structural tensions between environmental protection and economic interests. Some provinces, notably Quebec and British Columbia, tend to adopt more progressive climate policies, whereas others with resource-intensive economies are more cautious or resistant [9]. This combination of political decentralization, linguistic duality, and regional economic variation produces diverse policy approaches and, in turn, shapes how climate change is discussed in the media. Yet systematic analysis of climate communication in Canada and elsewhere has long been constrained by methodological obstacles: until recently, large-scale corpus collection and precise, reproducible content analysis were virtually impossible. Supplementary Table S1 provides an overview of prior studies in this area, highlighting their scope, focus, and coverage. The dominant pattern is one of corpora restricted to a few outlets, short observation windows, and English-language sources.

The CCF Database was designed to address four major limitations in previous research on climate change media coverage in Canada. First, the CCF draws on a larger set of news outlets and articles, comprising 266,271 articles from 20 newspapers with substantial representation for each outlet. Previous studies generally relied on smaller corpora and a limited number of newspapers, thereby capturing a narrower media landscape. Second, the CCF spans nearly five decades, providing a broader temporal perspective than most existing studies, which often focus on short periods or extend only from the late 1980s to the 2010s. Third, while climate attitudes vary significantly across regions [10], regional climate coverage remains understudied. Existing studies generally focus on nationally circulating newspapers—most commonly *The Globe and Mail*, *National Post*, and *Toronto Star*—and rarely compare a large set of provinces and territories. The CCF therefore includes a wide range of regional newspapers. Finally, the CCF reflects Canada’s linguistic diversity through a bilingual corpus that includes a substantial share of French-language content (82.9% English, 17.1% French), with francophone outlets both inside and outside Quebec. While Canada is officially bilingual, previous research has largely excluded French-language media from analysis. Two studies, in particular, have examined francophone coverage: Young and Dugas [11], comparing climate coverage between French-language outlets from Quebec and English-language outlets from the rest of Canada, and King et al. [12], analysing climate change’s public-health framing with partial inclusion of Quebec.

Methodologically, the CCF database is the first climate media corpus to offer sentence-level annotations derived from supervised machine learning and natural language processing (NLP). Unlike unsupervised approaches such as topic modelling, which remain dominant in the field, the CCF methodology relies on transformer-based classifiers (BERT for English, CamemBERT for French) trained on over 4,000 manually expert-coded sentences. NLP enables scholars to examine much larger corpora in far less time, opening new avenues for systematic inquiry [13, 14, 15]. However, as summarised in Supplementary Table S1, when climate researchers do employ NLP it is most often to identify key actors through named-entity recognition, to detect stance, or to conduct sentiment analysis, which tend only to map actors, tone, or evaluative direction rather than interpretive structures. The dominant computational approach in the field is still unsupervised topic modelling, where topics are treated as proxies for frames—an assumption that Nicholls and Culpepper [16] and Ot-makhova, Khanehzar, and Frermann [17] flag as problematic, as topics are too narrow to

capture the defining features of frames, which, following Entman [18], require identifying a problem, specifying causal interpretations, moral evaluations, and possible solutions. By contrast, the supervised approach adopted in the CCF Database enables the identification of frames, actor types, events, solution strategies, and emotional tone with a level of semantic precision that unsupervised methods cannot achieve. Only a few studies have applied comparable transformer-based models in this domain: Luo, Card, and Jurafsky [19] trained a single BERT classifier to analyse the evolution of climate partisanship in U.S. media; Meddeb et al. [20] used a CamemBERT model to detect climate misinformation in French news; Frermann et al. [21] fine-tuned transformer models to detect narrative roles (conflict, villain, resolution) in media articles on three issues including climate change; Badullovich et al. [22] benchmarked a RoBERTa classifier against zero-shot LLMs on issue and rhetorical frames in climate change editorials published in *Nature* and *Science*. None of these works, however, have trained and deployed a comparable suite of transformer models to systematically annotate climate-related media coverage at the scale, breadth, and bilingual coverage of the CCF database.

The CCF trained models attain a macro F1 score of 0.866 across all 65 annotation categories, as validated against an independent gold standard of 1,000 expert-coded sentences whose reliability was confirmed through double-coding (Cohen’s $\kappa = 0.596$, Krippendorff’s $\alpha = 0.698$, Gwet’s AC1 = 0.894, on the blind-coding phase of sentences 601–1000). In addition, named-entity recognition (NER) was applied to extract persons, organisations, and locations mentioned in the articles, and the entire corpus was embedded with the BAAI/bge-m3 multilingual encoder so that semantic-similarity queries can be issued directly against the deposited database. With over 60 BERT and CamemBERT classifiers covering frames, actors, events, solutions, and emotional tone, the CCF database fills a clear methodological gap by introducing a supervised NLP approach capable of detecting multiple analytical dimensions at scale, and the absence of raw text in the deposit is offset by the depth of the annotation: every sentence carries 65 binary labels, a deduplicated NER signature, and a 1024-dimensional embedding, so researchers can run frame-based, actor-based, and semantic queries directly on the deposited tables. This technical paper presents the methodology underlying the CCF database and its structure. The remainder of the document outlines the full workflow (data acquisition and preprocessing, model training, validation, and database enrichment) and offers the conceptual rationale and practical steps necessary for researchers wishing to work with the CCF database or to adapt its methodology elsewhere.

Methods

The CCF Database was built through a mixed-methods design that coupled large-scale data collection with state-of-the-art machine learning (ML). The resulting pipeline provides a systematic and reproducible framework for large-scale analysis of climate discourse that allows researchers to track how climate change is discussed, how narratives evolve over time, and who participates in the conversation.

Figure 1 presents the complete methodological pipeline and illustrates how the CCF Database was assembled from raw newspaper articles into a fully annotated database suitable for robust and reproducible climate communication research, both quantitative and

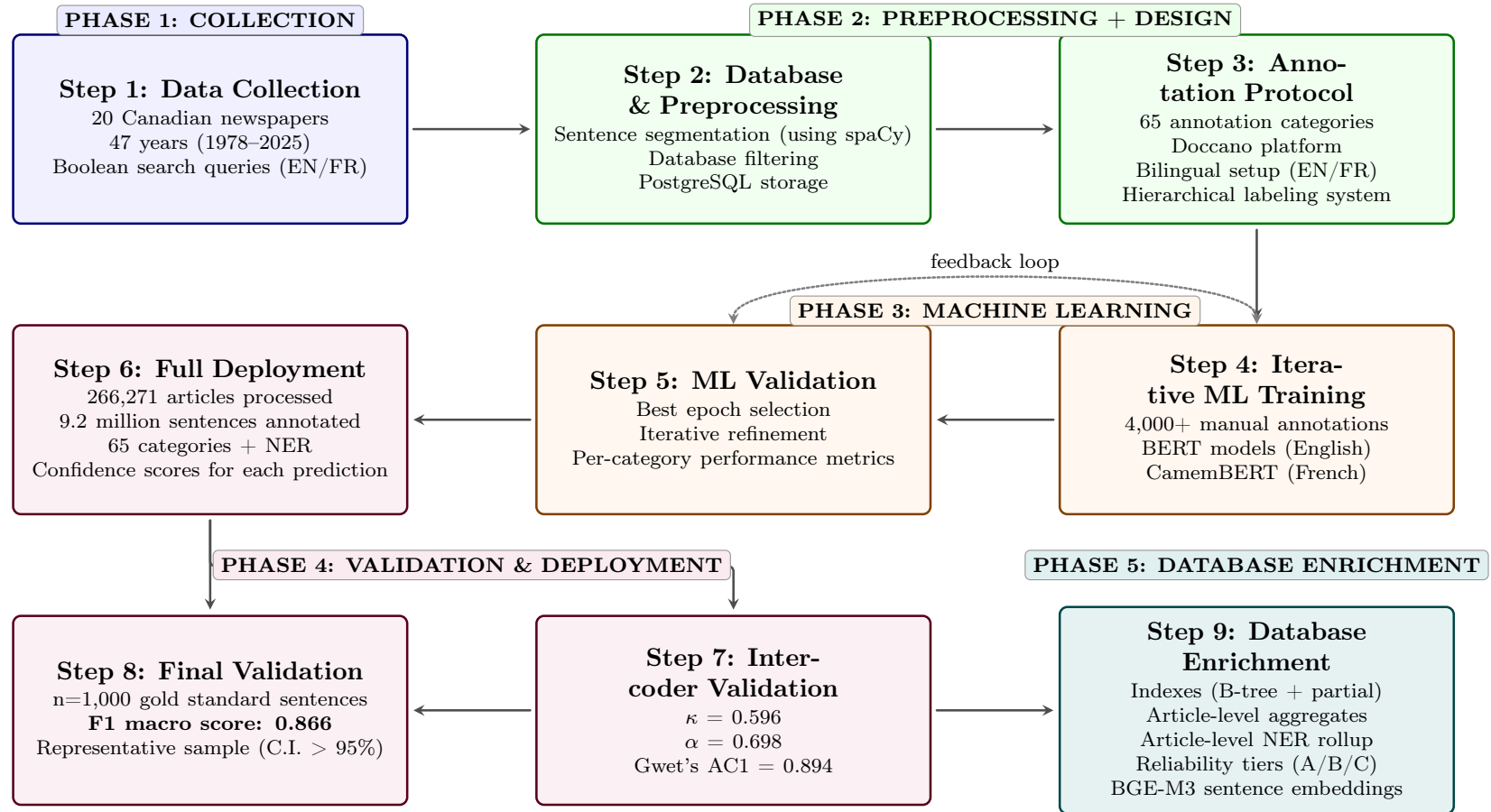


Figure 1: Methodological pipeline of the Canadian Climate Framing (CCF) project.

qualitative. The pipeline combined human expertise with computational methods through an iterative human-in-the-loop machine learning procedure. This approach ensured both the scalability needed to process 266,271 climate-related articles and the accuracy required for rigorous academic research.

The methodology comprises five distinct phases: (1) the *Collection Phase* involved systematic gathering of climate-related articles from 20 major Canadian newspapers using search queries resulting in 266,271 articles spanning 1978–2024; (2) the *Preprocessing and Design* phase transformed raw text into structured data, including sentence segmentation, language verification, and development of a comprehensive 65-category annotation framework; (3) the *Machine Learning* phase created and validated transformer-based machine learning models (BERT for English, CamemBERT for French) through iterative training with over 4,000 expert manual annotations; (4) the *Validation and Deployment* phase ensured rigorous quality assurance by combining standard performance metrics with a human evaluation protocol based on an expert-coded gold standard that was independently double-coded by a second annotator to assess intercoder reliability, before deploying the trained models to annotate the complete corpus of 9.2 million sentences; and (5) the *Database Enrichment* phase consolidated the sentence-level output into a publication-ready relational system by adding four derived tables (article-level annotation aggregates, article-level named-entity rollups, per-category reliability tiers, and 9.46 million BAAI/bge-m3 sentence-and-title embeddings indexed for approximate nearest-neighbour search), together with a complete index structure, so that the deposit can be queried directly without rewriting the same aggregation logic on every project.

Phase 1 & 2: Data collection, preprocessing, database creation and annotation protocol

Step 1: Data collection

The CCF database was built through a systematic protocol to trace the evolution of climate discourse in Canada’s bilingual print media, reaching back to the earliest available archives.

Articles were retrieved through university library access to three press databases: Factiva, Eureka.cc, and ProQuest Canadian Major Dailies. These platforms are the standard providers of full-text archives for Canadian newspapers and are routinely used in peer-reviewed media research [11, 12, 20]. Each platform was queried independently with the same bilingual Boolean keywords (see Table 1 below) restricted to the 20 selected outlets and to the full available time range, and duplicate articles appearing in more than one platform were collapsed at preprocessing time on the basis of newspaper, date, headline, and a 95% fuzzy similarity of the article body (see Step 2). Full bibliographic metadata are included in the deposit so that the corpus remains transparent and reproducible while staying within the bounds of copyrights. The design relies on the fair-dealing research exception in Canadian copyright law and the analogous fair-use research exception in U.S. law, both of which authorise non-consumptive, transformative use of copyrighted text for academic research; researchers with institutional access to Factiva, Eureka.cc, or ProQuest Canadian Major Dailies can recover the original text from the bibliographic coordinates.

A total of 20 newspapers were selected for the analysis (see Supplementary Table S2). The

objective was to include both national and regional outlets in order to ensure broad geographic representation. Within this pool, newspapers were primarily chosen based on circulation levels [23], with the highest-circulating outlets retained. Additional criteria also informed the selection, including the desire to capture a range of political orientations and to build a bilingual corpus that reflects Canada’s linguistic landscape. The corpus accordingly includes French-language outlets both inside and outside Quebec, together with one English-language outlet located in Quebec; this asymmetry mirrors the country’s linguistic geography, in which Quebec is the only province with French as its sole official language while New Brunswick is officially bilingual and several other provinces—most notably northeastern Ontario along the Quebec border—host sizeable francophone communities. The final selection was also partly shaped by availability constraints. When the highest-circulating newspaper of a given province was unavailable, the next highest-circulating outlet was retained instead; for Prince Edward Island, Nunavut, and the Northwest Territories, no eligible newspaper was available for inclusion at all.

The corpus spans 47 years, from 1978 to 2024, with 1978 marking the earliest article identified, quoted in the introduction of this paper. To identify climate-related articles, the authors used a transparent keyword-based filter with seven climate-domain terms per language (Table 1). To be included in the data set, an article had to contain at least one of these keywords in its title or body. These queries yielded an initial corpus of over 300,000 articles. As with any keyword-based retrieval, the procedure imposes a recall ceiling: it cannot capture paraphrastic, metonymic, or strictly context-only mentions of climate change, and the deposit therefore covers articles that explicitly use one of the query terms rather than the broader set of articles that discuss climate change in some form.

Table 1. Boolean queries used for article retrieval

Language	Boolean Query
English	"global warming" OR "climate change" OR "climate disruption" OR "climate disturbance" OR "climate disturbances" OR "climate crisis" OR "greenhouse gas"
French	"réchauffement climatique" OR "réchauffement planétaire" OR "changement climatique" OR "changements climatiques" OR "dérèglement climatique" OR "crise climatique" OR "gaz à effet de serre"

Step 2: Preprocessing

The preprocessing phase transformed the raw data into a structured dataset suitable for machine learning. Each article was segmented into two-sentence contexts using language-specific spaCy models—`en_core_web_lg` for English text and `fr_dep_news_trf` for French text, both optimised for news content. Sliding windows with single-sentence overlaps were implemented to ensure complete coverage while maintaining semantic continuity throughout each article. This level of granularity supported machine-learning applications by increasing the size of the training dataset. More importantly, working at the (near-)sentence level was the only approach that allowed ex-post analysis to remain highly granular compared to paragraphs, where multiple frames, actors, or solutions can co-occur and become indistinguishable. Article-level variables could then be reconstructed with high granularity by

aggregating sentence-level annotations (e.g., percentage of sentences in an article that mention a specific frame or actor), whereas paragraph-level annotations would have been less precise and more difficult to interpret.

Quality control measures were applied systematically throughout the data preparation pipeline. Duplicate articles were identified and removed with fuzzy string-similarity algorithms set to a 95% threshold. Any article containing less than 100 words were also removed to eliminate reader letters, which often did not provide enough material to identify the presence of a frame and did not qualify as a press article per se. However, all other types of text (opinion columns, letters to the editor, news articles, editorials, etc.) were retained. Regarding this matter, Lawrence [24] explains that it is precisely in editorials, or more generally in opinion texts, where journalistic standards are applied with a greater degree of flexibility, that the battle of framing is best observed.

Articles published by external news agencies (for example, *Agence France-Presse*, *Can-West News Service*, *Reuters*, or *Bloomberg*) or by other newspapers not included in the analysis but which often belong to the same media conglomerate as the selected outlets (for example, *the Financial Post*) were also retained. These republications were treated as aligned with the editorial direction of the host news outlet, and a republication did not contradict the objectives of the study since it indicated editorial interest in a topic concerning climate change. This process reduced the corpus to 266,271 unique articles. Duplicates of the same article appearing in different media outlets were deliberately kept, since their co-occurrence is itself an object of analysis when measuring media bubbles. Language was then verified with the pre-trained fastText language-identification model [25, 26], in order to ensure consistent categorisation of bilingual content, and publication dates were standardised at the same step. This data preparation produced a clean, standardised corpus ready for machine learning and annotation, stored in two PostgreSQL tables: one with full metadata (titles, main text, authors, dates, and page numbers) and another with the processed sentence segments and their corresponding identifiers.

The resulting database is illustrated in Figure 2, which displays the distribution of article counts by media (20 outlets), while Figure 3 shows the total article counts per year across the full time span. Data extraction concluded in February 2025, and 2025 was therefore excluded from the yearly counts to avoid reporting on an incomplete year; the authors plan to update the database periodically and develop it into an observatory of Canadian climate discourse. Finally, Figure 4 presents the distribution of articles by province, based on the primary location of each newspaper. This geographic breakdown shows the representativeness of the database across Canada, with 82.9% of articles in English and 17.1% in French. Three national newspapers—*The Globe and Mail*, the *National Post*, and the *Toronto Star*—are omitted from the provincial breakdown because of their pan-Canadian scope; nevertheless, they account for 36.2% of all articles.

Step 3: Annotation protocol

The annotation framework constitutes the conceptual core of the CCF database. It comprises 65 categories organised into a hierarchical taxonomy designed to capture multiple dimensions of climate discourse at near-sentence granularity. The framework was developed through a combined theory-driven and data-driven procedure: it drew on established scholarship in

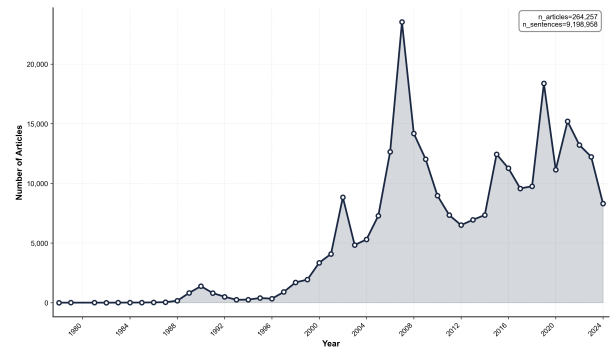
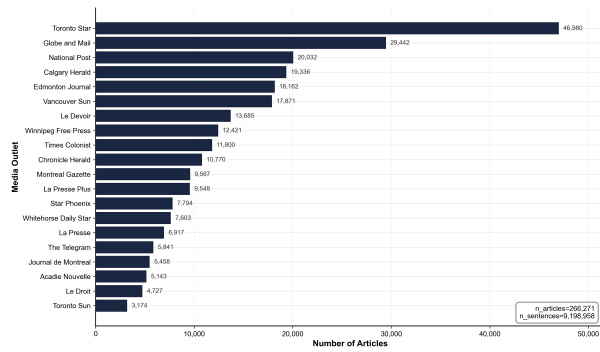


Figure 2: Distribution of articles by media.

Figure 3: Total number of articles per year.

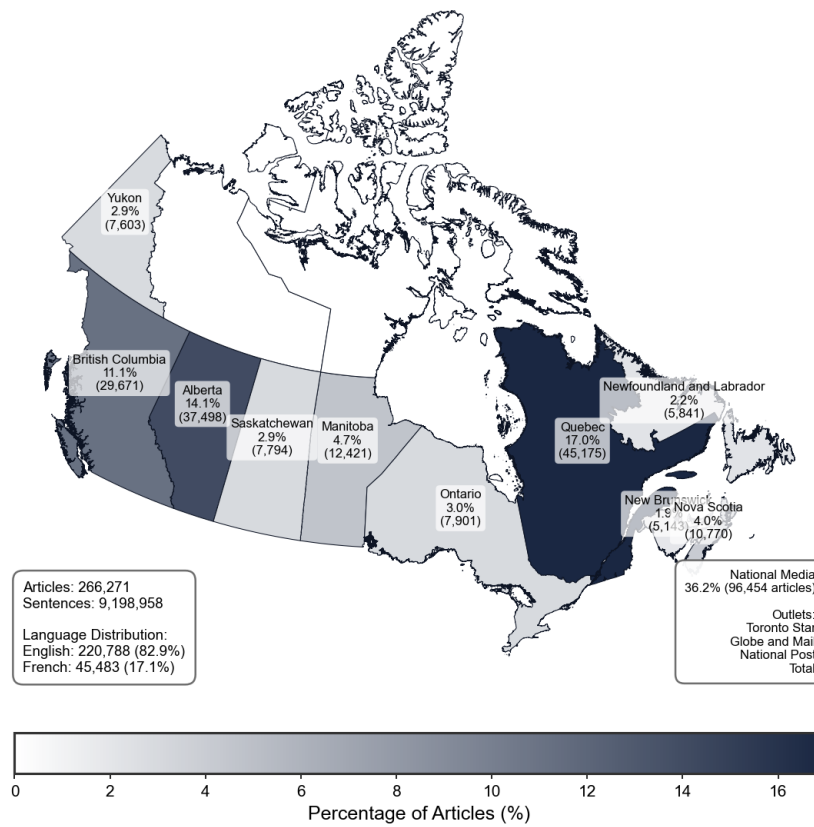


Figure 4: Distribution of articles by province.

framing, agenda-setting, climate communication, and social problems studies, and was refined inductively through iterative engagement with the corpus and extensive pretesting. The full taxonomy and operational definitions are reported in Supplementary Table S3, while Table 2 provides a compact overview of the framework’s structure. The framework was organised around *Primary Categories* corresponding to broad analytical dimensions, and *Sub-categories* capturing internal distinctions within each *Primary Category* (Table 2).

Primary categories. The framework was organised around four primary analytic categories that structure the annotation protocol: *Main Frames*, *Actors/Messengers*, *Events*, and *Solutions* (Table 2). Drawing on climate communication research, these categories acknowledge that media coverage does more than describe the issue, since it also selects the interpretive lens through which it is read and understood, the actors who are granted voice, the events that anchor attention, and the courses of action that are put forward as plausible.

First, for *Main Frames*, the authors built on the general framing tradition often associated with Entman [18], which encompasses four dimensions: problem definition, causal interpretation, moral evaluation, and treatment recommendations. In practice, however, applying this framework to climate discourse presents significant challenges. Individual statements frequently span multiple dimensions, and "treatments" (i.e., proposed policy responses) proved particularly difficult to identify consistently [27]. This difficulty was amplified by the nature of climate change as a “wicked problem,” where proposed solutions remain subject to scientific uncertainty and political contestation [28]. For these reasons, many studies do not apply Entman’s four dimensions strictly [29]. The CCF framework addressed these challenges by operationalising main frames around major societal domains in which climate change is repeatedly interpreted in media discourse: economic, security, public health, cultural, environmental, political, scientific, and justice spheres (Table 2 and Supplementary Table S3). This domain-based approach retained the core elements of problem definition, causal interpretation, moral evaluation, and treatment recommendations where they naturally emerged in the data.

Second, the framework treated *Solutions* as a distinct primary family rather than embedding them inside each main frame. This choice reflected both the empirical ambiguity of policy prescriptions in climate coverage and the substantive interest in distinguishing mitigation from adaptation, the latter having historically received less media attention [30]. Third, the *Actors/Messengers* family was grounded in the long-standing finding that *who* speaks about climate change matters for credibility, salience, and persuasion [31, 32]. Finally, the *Events* family reflected agenda-setting scholarship showing that media attention is cyclical and often driven by external shocks and institutional moments (e.g., disasters, major conferences, reports) [33, 34]. Together, these four primary categories provided a theory-informed structure for capturing how climate change is framed, who is authorised to speak, what triggers attention, and which actions are articulated (Table 2 and Supplementary Table S3).

Operationalisation: hierarchical taxonomy and sub-categories. Operationally, the framework implemented a hierarchical labelling logic that mirrors its conceptual structure (Table 2 and Supplementary Table S3). Each *Primary Category* contains (i) a detection-level category that captures whether the broader dimension is present in a two-sentence unit, and (ii) fine-grained sub-categories that specify which internal distinctions apply when the primary category is detected. This design enabled multi-label annotation: multiple frames, actors, events, and solutions could co-occur within the same unit, and the same applied to

sub-categories. Within *Main Frames*, each domain frame (e.g., Economic, Health, Justice) was paired with sub-categories capturing recurring interpretive distinctions observed in the literature and in the corpus (Table 2 and Supplementary Table S3). Importantly, these sub-categories were not treated as mutually exclusive: a unit could simultaneously describe, for instance, the negative economic impacts of climate change and the economic benefits of climate action. The same principle applied to *Actors/Messengers*, where a single person could legitimately be coded as multiple expert types (e.g., a physician would be classified as having both medical and natural science expertise). For full category definitions, see Supplementary

Table 2: The CCF annotation framework: 65 hierarchical categories for comprehensive climate discourse analysis

Main Category	N ^a	Main category detection (<i>in italics</i>) and sub-categories
Primary Categories: Main Frames (8 frames and their sub-categories)		
Economic	6	<i>Economic frame</i> , negative/positive impacts on the economy, economic disadvantages/benefits of climate action, carbon footprint of the economic sector
Health	3	<i>Health frame</i> , negative impacts on health, health co-benefits of climate action
Security	5	<i>Security frame</i> , climate refugees, conflict, post-disaster military assistance, disruption of military operations
Justice	6	<i>Justice frame</i> , winners/losers of climate action, differentiated responsibility, unequal vulnerability, unequal access to climate action, intergenerational justice
Political	5	<i>Political frame</i> , policy action, political debate, political positioning, public opinion data
Scientific	5	<i>Scientific frame</i> , scientific debate, popularisation or discovery, questioning of climate science, defense of climate science
Environmental	3	<i>Environmental frame</i> , loss of natural environments, loss of fauna and flora
Cultural	5	<i>Cultural frame</i> , artistic representation, disruption of cultural or sports events, loss of indigenous practices, carbon footprint of the cultural and sports sectors
Primary Categories: Actors, Events and Solutions (and their sub-categories)		
Actors/Messengers	9	<i>Any messenger</i> , scientists, politicians, activists, medical/economic/security/legal/cultural experts
Events	6	<i>Any event</i> , natural disasters, conferences, reports, elections, policy announcements
Solutions	3	<i>Any solution</i> , mitigation, adaptation
Emotional Tone	3	Positive (hope, optimism), negative (fear, anxiety), neutral/none
Geographic Focus	1	Canadian places, actors, data, and policies
Urgency to act	1	Conveys immediate danger, crisis, or “code red” urgency
Named Entities	3	Persons (PER), organizations (ORG), locations (LOC)
Total: 65 categories		

^aN counts the main category detection (shown in italics) together with its sub-categories; e.g., the Economic frame: 1 main category detection + 5 sub-categories = 6. Flat categories (Emotional Tone, Named Entities) and single-detection categories (Geographic Focus, Urgency to act) have no separate main category detection.

Table S3.

Other dimensions: context, tone, urgency, and entity extraction. In addition to the four *Primary Categories*, the framework included theoretically motivated dimensions that capture context and meaning-making beyond framing in the strict sense (Table 2 and Supplementary Table S3 and S11). A *Geographic Focus* category recorded whether coverage situated climate change in a Canadian context, reflecting concerns that climate change is often portrayed as geographically distant [35]. An *Emotional Tone* category captured positive, negative, or neutral valence, consistent with research showing that emotions shape how audiences interpret and respond to climate discourse [36, 37]. An *Urgency to act* category captured alarmist or “code red” signals that communicate immediacy and crisis framing. Finally, Named Entity Recognition (NER) extracted and classified *Persons*, *Organizations*, and *Locations* into a JSON field, enabling downstream analyses such as cocitation networks and the mapping of epistemic authorities (Table 2; Supplementary Table S3).

Pretesting and iterative refinement. Finally, alongside these theoretical design choices, the framework was iteratively refined through inductive coding rounds. As sentences were annotated, emergent themes sometimes prompted targeted additions and clarifications to category definitions, followed by pretesting to ensure internal consistency and coder interpretability. This procedure strengthened the framework’s construct validity and ensured that the final taxonomy remained both theoretically anchored and empirically adequate. The following real example from the database—an excerpt from a 2022 article—illustrates the corresponding manual annotation protocol:

Example (annotated sentence).

“In Scientific American, professor of Environmental Studies at Humboldt State University Sarah Jaquette Ray writes, ‘Climate anxiety can operate like white fragility, sucking up all the oxygen in the room and devoting resources toward appeasing the dominant group.’ But it would be a mistake to conclude that this term isn’t applicable in the Global South, period.”

Tagged Dimensions	Tagged categories
Actors/Messengers	Scientists
Main frames	Justice (primary category) + North–South responsibility; Health (primary category)
Emotional tone	Negative

Phase 3: Machine learning

The development of the training data followed an iterative manual annotation protocol aligned with the annotation framework previously described. A single expert annotator (climate policy and communication) labelled sentences in sequential batches (first 1,000 sentences, then three additional batches of 1,000 each) for a total of 4,000 annotated samples. The choice of a single highly-trained expert during training, with an independent second coder reserved for the gold-standard validation, was made to maximise definitional consistency across the 4,000 training sentences. In framing analysis, where 65 conceptually overlapping categories must be applied to fine-grained sentence units, multi-coder training can introduce drift in operational definitions that exceeds the noise it removes; the more reliable

protocol is therefore to stabilise the codebook with a single expert and to validate it against an independent coder on a held-out gold standard [38]. The batching served two explicit objectives that were only attainable with a single coder: (1) to iteratively improve and stabilise the annotation guidelines after each round; and (2) to progressively monitor and optimise model training to reach the highest attainable F1 score. This process culminated in a macro F1 score of 0.826 (see Table 3) during the training phase (see Supplementary Table S4 and S11 for detailed metrics).

The sampling process drew equally from English and French language groups, with final annotation counts reaching 1,927 English and 2,073 French sentences. The training dataset was subsequently partitioned using stratified random sampling to create training (80%) and validation (20%) databases for each category and language. Supplementary Table S5 provides the complete distribution of training and validation samples for all annotation categories. The random sampling approach ensured broad coverage across the temporal span and diverse media sources. The annotation protocol was developed with detailed operational definitions for each of the 65 categories to ensure consistency.

The machine learning pipeline leveraged transformer architectures optimised for each language through a custom fork of the AugmentedSocialScientist library [39, 40]. The fork built from Do, Ollion, and Shen [39] added several functionalities that were central to ensuring robust machine-learning training (metric logging at every training epoch, best-model selection using a weighted F1 score ($0.7 \times F_1^{(1)} + 0.3 \times \text{macro-}F_1$), and an automated reinforced training protocol for underperforming models, described below) [40]. English text processing employed BERT-base-uncased, while French text used CamemBERT-base, both containing 110 million parameters. The training strategy implemented hierarchical classification, where detection models were trained first as binary classifiers on the full annotated dataset, followed by sub-classification models trained exclusively on manually annotated sentences that were positive for the corresponding primary category. This approach minimised false-positive propagation and allowed for specialised optimisation for each classification task.

Models failing to achieve a positive-class F1 score of at least 0.60 during the training phase underwent an automated reinforced training protocol. This protocol was triggered for 45 of the 128 trained models (35%), concentrated in low-prevalence sub-frames—cultural, justice, and security—and in the fine-grained political and scientific sub-categories. The reinforced phase implemented weighted random sampling to oversample minority classes, doubled the batch size to 64, reduced the learning rate to 1×10^{-5} , and applied a weighted cross-entropy loss with emphasis on the positive class. Training was extended for an additional 20 epochs, after which the best epochs of the normal and reinforced phases were compared on a combined criterion that gave 70% weight to the positive-class F1 and 30% to the macro F1, i.e. $0.7 \cdot F_1^{(1)} + 0.3 \cdot \text{macro-}F_1$. The reinforced epoch was retained as the final model for 23 of these 45 cases, while the remaining 22 reverted to their best normal-training epoch; averaged over the 23 reinforced models eventually retained, the positive-class F1 improved by +0.25, lifting most of the affected categories above the deployment threshold. Five categories could nevertheless not be trained at all because of insufficient positive annotations in the training data, and were consequently excluded from the annotation framework: *carbon footprint of the health sector*, *positive impacts of climate change on health*, *post-disaster military assistance*, *disruption of military operations due to climate change*, and *carbon footprint of the defense sector*. The complete per-model training provenance, including which models triggered reinforcement and

which retained the reinforced epoch, is provided in Supplementary Table S11.

Phase 4: Validation and deployment

Phase 4 chained the full deployment of the trained models (Step 6) with the two validation steps (Steps 7 and 8) that closed the methodological pipeline. Step 6 is described immediately below; the design of the validation steps (sampling protocol, gold-standard construction, model performance metrics, and inter-coder reliability) is described separately and in full in the [Technical Validation section](#) below.

Step 6: Full deployment

Once the per-category models had been finalised, all trained models were applied to annotate the complete CCF Database of 9.2 million sentences across 266,271 articles. A hierarchical classification strategy was implemented that directly mirrored the annotation protocol’s structure of primary categories and their associated sub-categories, as illustrated in Figure 5.

The hierarchical approach reflected the conceptual organisation of the annotation framework with eleven primary detection categories, each with specific sub-categories. The eight main frames (*Economic, Health, Security, Justice, Political, Scientific, Environmental, and Cultural Frame*) functioned as primary categories with their internal distinctions—for example, *Economic Frame*, when positive (=1), triggered five economic sub-models (*nega-*

Table 3: Model training performance metrics for primary annotation categories

Category	F1 (Class 1)		F1 (Class 0)		Macro F1	
	EN	FR	EN	FR	EN	FR
<i>Main Frames</i>						
Economic Frame	0.745	0.814	0.944	0.957	0.845	0.885
Health Frame	0.800	0.667	0.989	0.994	0.894	0.830
Security Frame	0.870	0.800	0.996	0.997	0.933	0.898
Justice Frame	0.719	0.717	0.975	0.981	0.847	0.849
Political Frame	0.808	0.774	0.897	0.888	0.853	0.831
Scientific Frame	0.784	0.702	0.953	0.962	0.869	0.832
Environmental Frame	0.842	0.625	0.989	0.980	0.915	0.802
Cultural Frame	0.773	0.833	0.986	0.993	0.879	0.913
<i>Other Primary Categories: Actors, Events and Solutions</i>						
Presence of Messengers	0.912	0.929	0.904	0.915	0.908	0.922
Presence of Events	0.794	0.819	0.935	0.932	0.865	0.876
Presence of Solutions	0.737	0.878	0.914	0.944	0.825	0.911
Mention of Canada	0.942	0.964	0.968	0.980	0.955	0.972
Urgency to act	0.591	0.649	0.975	0.984	0.783	0.816
Overall Average*	0.794	0.782	0.956	0.962	0.875	0.872
Combined (ALL)	0.753		0.900		0.826	

*Overall average across all active annotation categories (see Supplementary Table S4 for complete metrics).

tive/positive economic impacts of climate change, economic disadvantages/benefits of climate action, carbon footprint of the economic sector), while a negative detection (=0) bypassed these entirely.

Similarly, the three other primary categories followed this conditional logic: *Actors/Messengers Detection* activated nine actor-type sub-models (*Health Expert, Economic Expert, Security Expert, Legal Expert, Cultural Expert, Natural Scientist, Social Scientist, Activist, Public Official*), *Event Detection* triggered eight event-type classifications (*Extreme Weather Event, Meeting/Conference, Publication, Election, Policy Announcement, Judiciary Decision, Cultural Event, Protest*), and *Solutions Detection* applied two solution sub-categories (*Mitigation Strategy, Adaptation Strategy*). The deployment also processed standalone primary categories without hierarchical structure—*Emotional Tone* (*Positive Emotion/Negative Emotion/Neutral Emotion*), *Geographic Focus* (*Canadian Context*), *Urgency/Alarmism*.

In addition to the custom-trained classification models, three pre-trained Named Entity Recognition (NER) models were employed, selected on the basis of empirical evaluation on the CCF data. A hybrid approach tailored to each language was implemented. For English texts, the publicly available `dslim/bert-base-NER` model (hosted on the Hugging Face Hub) was used for all three entity types: persons, organisations, and locations. For French texts, person entities were extracted with the `spaCy fr_core_news_lg` pipeline, while organisations and locations were extracted with the `Jean-Baptiste/camembert-ner` model (also hosted on the Hugging Face Hub). This hybrid strategy was chosen to leverage the respective strengths of each model: spaCy’s superior performance on French person names, and CamemBERT-NER’s robust identification of French organisational and geographical entities. Per-language

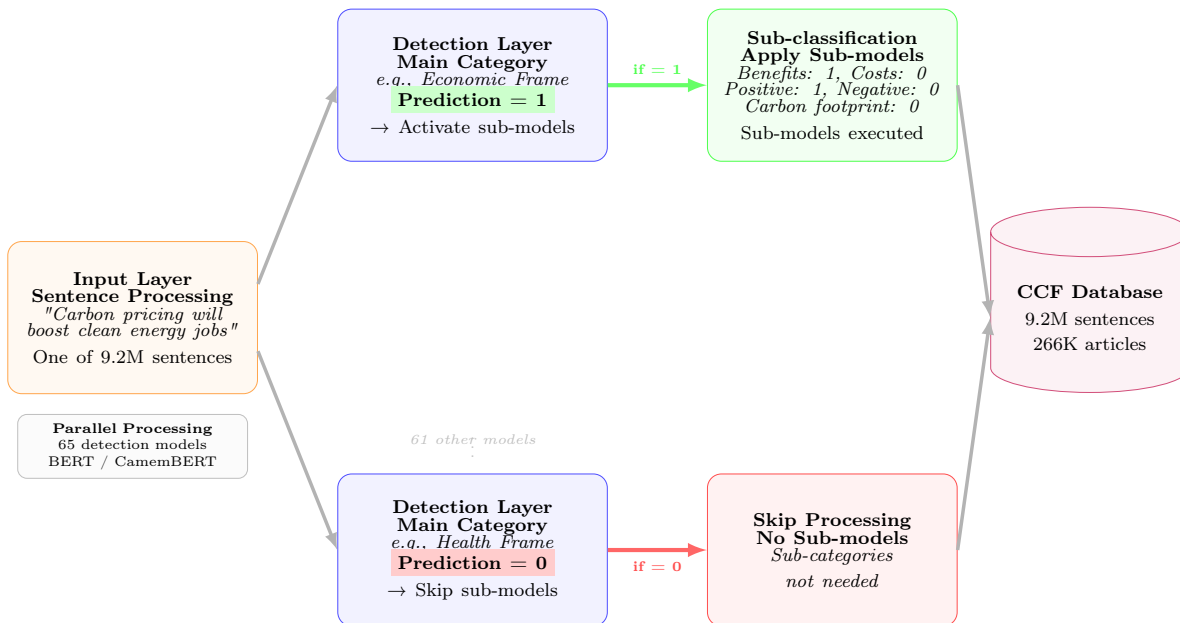


Figure 5: Hierarchical annotation pipeline. The system processes each sentence through detection models that determine whether corresponding sub-category models should be applied, enabling efficient annotation of the entire corpus.

performance metrics for the three NER models are reported in Supplementary Table S6.

Phase 5: Database enrichment

The deployment described above produced the two original tables of the CCF Database (`CCF_full_data` and `CCF_processed_data`). The database was then further enriched with four additional tables and a complete index structure so that researchers could interrogate the corpus without rewriting the same article-level aggregations on every project. The enrichment procedure was deterministic and idempotent: it ran end-to-end from a freshly restored database dump, and each step (index provisioning, article-level aggregates, article-level entities, sentence embeddings, and reliability tiers) is described in detail below.

Indexes. The first enrichment step provisioned twenty indexes on the two original tables: unique primary keys on (`doc_id`) for `CCF_full_data` and (`doc_id`, `sentence_id`) for `CCF_processed_data`; routine B-trees on the columns researchers query most frequently (`date`, `media`, `language`); and partial B-trees on `doc_id` restricted to the rows where one of the eleven primary detection flags is positive (*economic_frame*, *health_frame*, *security_frame*, *justice_frame*, *political_frame*, *scientific_frame*, *environmental_frame*, *cultural_frame*, *messenger*, *event*, *solution*). The partial indexes accelerated the most common analytical workflow—retrieving the subset of articles or sentences in which a given frame is positive—without paying the storage cost of a full B-tree on each binary column.

Article-level aggregates. The `CCF_article_aggregates` table was built with one row per `doc_id`; for each of the 65 annotation categories, it stored the article-level proportion (`prop_X`, in $[0, 1]$) of sentences positive for category *X*, computed as the sum of the binary flag divided by the article’s sentence count. The table also recorded the article’s sentence count overall and by language (`n_sentences`, `n_sentences_en`, `n_sentences_fr`), the article’s top frame (`top_frame`, the main frame with the highest proportion in the article), the proportion of that frame (`top_frame_prop`), the Shannon entropy of the eight main-frame proportions (`entropy_frames`, measuring how concentrated or diffuse the article’s framing is), and the number of distinct messenger, event, and solution sub-categories present in the article (`n_unique_messenger_subs`, `n_unique_event_subs`, `n_unique_solution_subs`). Sub-categories whose parent primary detection was negative were coalesced to zero, so a `prop_X` reading of 0 unambiguously means “no sentence in this article is positive for category *X*”, regardless of whether the sub-category was hierarchically evaluated.

Article-level named entities. The `CCF_article_entities` table consolidated the sentence-level Named Entity Recognition output (the `ner_entities` JSON column in `CCF_processed_data`) into one row per article. Three JSONB arrays (`entities_per`, `entities_org`, `entities_loc`) stored the deduplicated set of persons, organisations, and locations mentioned anywhere in the article; three counters (`n_unique_per`, `n_unique_org`, `n_unique_loc`) recorded their cardinalities; and three further counters reported how many sentences of the article contained at least one entity of each type. The presence of an entity in this rollup does not by itself disambiguate whether the entity is the speaker, the subject, or a passing reference; users wishing to recover speaker–statement attribution should combine the entity rollup with the sentence-level `messenger` family of annotations or with downstream attribution models that are out of scope for this database.

Sentence embeddings. The `CCF_sentence_embeddings` table was populated, for every sentence in `CCF_processed_data` and for the title of every article (encoded as `sentence_id = 0`), with the dense vector representation produced by the BAAI `bge-m3` multilingual encoder [41]. `bge-m3` was selected for three reasons: (i) it natively supports the 100+ languages relevant to bilingual Canadian coverage and is reported to perform competitively on French–English cross-lingual tasks; (ii) it produces 1024-dimensional dense vectors that combine acceptable retrieval quality with manageable storage overhead; and (iii) it has been stable and widely adopted since its January 2024 release, which is favourable for the long-term reproducibility expected of a published dataset. Vectors were L2-normalised at encoding time, so cosine similarity reduces to a dot product and can be queried directly through the `pgvector <=>` operator. Storage used the `halfvec(1024)` type (2 bytes per dimension, identical to the upstream `float16` representation), and an HNSW index on the cosine operator class was built to deliver sub-second approximate top- k retrieval over the 9.2 million-row table.

Reliability tiers. The last enrichment step recorded, for each annotation category, a recommendation on how confidently it can be used in downstream analyses. The recommendation was summarised as a discrete *reliability tier* derived from the two evaluation criteria reported in full in the [Technical Validation section](#) below—the validation macro F_1 score against the gold standard, and the blind-phase Cohen’s κ between the expert annotator and the independent second coder. The `CCF_reliability_tiers` table was built as a 65-row lookup classifying each category into one of three tiers: Tier A (*research-grade*) required $F_1 \geq 0.80$ and $\kappa \geq 0.60$; Tier C (*exploratory*) corresponded to $F_1 < 0.60$ or $\kappa < 0.40$; Tier B (*use with caution*) captured everything in between. The table also flagged the categories affected by *prevalence-induced reliability deflation*—that is, categories for which $F_1 \geq 0.70$ but $\kappa < 0.40$, where the chance-corrected agreement is depressed by very low prevalence rather than by systematic disagreement between coders [42, 43]—and the categories that could not be trained at all in either language. Filtering downstream analyses to `tier_overall = 'A'` (or, equivalently, joining `CCF_processed_data` on `CCF_reliability_tiers.code`) restricts the analysis to the 27 categories that the authors recommend without reservation; the full tier assignment by language is provided in Supplementary Table S10.

A diagram of the enriched database, together with the join paths that connect the five tables, appears in the Data Records section (Figure 6). The complete enrichment procedure ran end-to-end in approximately 2.5 hours on a single Apple Mac Studio with the M2 Ultra chip and 128 GB of unified memory.

Data Records

Overview of the deposit

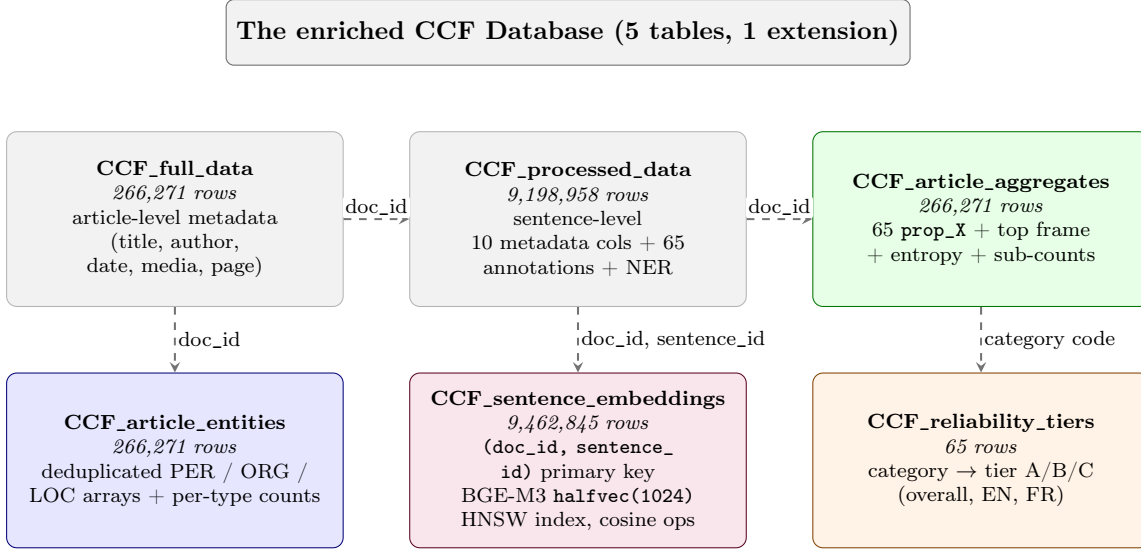


Figure 6: Schema of the enriched CCF Database. Dashed arrows indicate the join paths between tables; column types and full schemas are documented in Supplementary Table S12. `CCF_full_data` and `CCF_processed_data` are the original sentence-level resource; the three coloured groups are the article-level rollups (green), the reliability lookup (orange), and the embedding table (purple).

The CCF Database is deposited on Zenodo under a CC-BY 4.0 licence in two complementary editions sharing identical schemas. The canonical PostgreSQL edition (<https://doi.org/10.5281/zenodo.20667151> [44]) ships as a single tarball (`CCF_Database.tar`, ≈ 37 GB) wrapping a `pg_dump directory` archive of the six relational tables together with all B-tree, partial, and HNSW indexes; restoration requires PostgreSQL 16 or 17 with the `pgvector` extension ($\geq 0.8.2$ for `halfvec` support) and the deposit’s README provides the exact three-line restore command (`tar -xf, createdb, pg_restore -j 8`). A sister column-oriented Apache Parquet mirror (<https://doi.org/10.5281/zenodo.20667154> [45], one ZSTD-compressed file per table, ≈ 15.8 GB) is provided for users without a PostgreSQL backend; the `halfvec(1024)` embedding column is materialised as a 1024-element `LIST<FLOAT>` and the JSONB entity arrays are serialised as UTF-8 JSON strings, so each Parquet file can be read directly with `pandas`, `polars`, R/`arrow`, DuckDB, or Spark. The companion code, documentation, and reproducible pipeline—including the annotation pipeline used to build the database (`Scripts/Annotation/`), the canonical CSVs underlying the supplementary tables (`Database/Training_data/`), and the full training data in JSONL format—are bundled with both Zenodo editions as `ccf_code_and_paper.tar.gz` and additionally archived on the Open Science Framework at <https://doi.org/10.17605/OSF.IO/Q5W47> [46], with a regularly updated development mirror on GitHub at <https://github.com/antoinelemor/CCF-canadian-climate-framing>.

Because Canadian newspaper articles are protected by copyrights, the deposit does not redistribute raw sentence text or full article bodies. What is shared is everything else: article-level metadata (titles, authors, publication dates, newspapers, page numbers), sentence-level identifiers, the 65 binary annotations, the named-entity extractions, the article-level aggregates and entity rollups, the per-category reliability tiers, and the BGE-M3 sentence embeddings. Each article is identified by its bibliographic coordinates (`media`, `date`, `title`, `author`, `page_number`), which is the standard bibliographic reference for Canadian newspapers and is sufficient for any researcher with institutional access to Factiva, Eureka.cc, or ProQuest Canadian Major Dailies to retrieve the original text. No DOIs exist for the historical articles covered by the corpus.

The database is organised around two original tables produced by the annotation pipeline (`CCF_full_data`, `CCF_processed_data`) and four enrichment tables that pre-compute the queries most frequently issued by analysts (article-level aggregates, article-level entities, reliability tiers, sentence embeddings). The schema is summarised in Table 4 and depicted graphically in Figure 6; the join paths in that diagram point to the recommended foreign keys whenever a query needs to combine two tables.

Table 4: Overview of the six tables in the deposited CCF Database. The `pgvector` extension underpins the `CCF_sentence_embeddings` table.

Table	Rows	Granularity	What it stores
<code>CCF_full_data</code>	266,271	Article	Bibliographic metadata for every article in the corpus.
<code>CCF_processed_data</code>	9,198,958	Sentence	The 65 sentence-level binary annotations and the per-sentence NER output.
<code>CCF_article_aggregates</code>	266,271	Article	Per-article proportions, top frame, framing entropy and sub-category counts.
<code>CCF_article_entities</code>	266,271	Article	Deduplicated JSONB arrays of PER / ORG / LOC mentions, with per-type cardinalities and per-type sentence counts.
<code>CCF_reliability_tiers</code>	65	Category	Per-category reliability tier (A / B / C) with the underlying F_1 and κ values.
<code>CCF_sentence_embeddings</code>	9,462,845	Sentence + title	L2-normalised BGE-M3 1024-dimensional vectors, queryable through <code>pgvector</code> .

CCF_full_data – article-level metadata

This table is the canonical inventory of articles in the corpus and the natural entry point for any analysis that operates at the article level. Each of its 266,271 rows describes one unique article and is uniquely keyed by `doc_id`; every other table in the database joins back to `CCF_full_data` through this identifier. The columns are summarised in Table 5.

Table 5: Columns of `CCF_full_data`.

Column	Type	Definition
<code>doc_id</code>	BIGINT (PK)	Unique article identifier; foreign-key target for every other table.
<code>news_type</code>	TEXT	Editorial category: news, opinion, editorial, etc.
<code>title</code>	TEXT	Article headline.
<code>author</code>	TEXT	Author byline when available.
<code>media</code>	TEXT	Newspaper name.
<code>words_count</code>	BIGINT	Total word count of the article.
<code>date</code>	DATE	Publication date.
<code>language</code>	TEXT	“EN” or “FR”.
<code>page_number</code>	TEXT	Print page (free text, e.g. “A1” or “B3”).

CCF_processed_data – sentence-level annotations

This is the workhorse table of the database. It contains one row per two-sentence analytical unit, so 9,198,958 rows in total, and every annotation produced by the machine-learning pipeline lives here. The same ten metadata columns as `CCF_full_data` are denormalised on every row so that sentence-level queries can run without joining back to the article table whenever the analyst only needs the date, the newspaper, or the language. On top of that, the table stores the 65 binary annotation columns named exactly as in Supplementary Table S3, and one JSON column (`ner_entities`) holding the per-sentence Named Entity Recognition output. Sub-category columns are NULL when their parent primary detection is negative; researchers who want to treat “absent” as equivalent to “not detected” should coalesce NULL to zero, which is exactly the convention adopted by `CCF_article_aggregates`. The structural columns are summarised in Table 6.

Table 6: Structural columns of `CCF_processed_data`. The 65 binary annotation columns are defined in Supplementary Table S3.

Column	Type	Definition
<code>doc_id</code>	BIGINT	Foreign key to <code>CCF_full_data</code> .
<code>sentence_id</code>	BIGINT	Position of the sentence within its article (starting at 1). <code>sentence_id</code> = 0 is reserved for the article title in the embeddings table.
(metadata)	various	Ten metadata columns duplicated from <code>CCF_full_data</code> for query convenience: <code>news_type</code> , <code>title</code> , <code>author</code> , <code>media</code> , <code>words_count</code> , <code>date</code> , <code>language</code> , <code>page_number</code> .
65 annotation columns	INTEGER (0/1, nullable)	One binary flag per category from Supplementary Table S3. Sub-categories are NULL when the parent primary detection is negative.
<code>ner_entities</code>	TEXT (JSON)	Per-sentence NER output in the format {"PER": [...], "ORG": [...], "LOC": [...]}.

CCF_article_aggregates – article-level rollup

`CCF_article_aggregates` is the recommended starting point for most article-level analyses. It collapses the 9.2 million sentence-level rows into one row per article (266,271 rows) and pre-computes the quantities that almost every project needs: the proportion of sentences positive for each of the 65 categories, the top frame, a measure of how concentrated or diffuse the framing of the article is, and the number of distinct sub-categories activated. Because these aggregates are computed once at enrichment time and indexed, a sentence-level `GROUP BY` on the 9.2-million-row sentence table is rarely necessary once the aggregates are available. The columns are summarised in Table 7.

Table 7: Columns of `CCF_article_aggregates`.

Column	Type	Definition
<code>doc_id</code>	BIGINT (PK)	Foreign key to <code>CCF_full_data</code> .
<code>n_sentences</code>	INTEGER	Total number of two-sentence analytical units in the article.
<code>n_sentences_en, n_sentences_fr</code>	INTEGER	Same count restricted to English / French sentences.
65 <code>prop_X</code> columns	DOUBLE PRECISION ([0, 1])	Article-level proportion of sentences positive for category <i>X</i> . Sub-categories with a negative parent detection are coalesced to zero.
<code>top_frame</code>	TEXT	Main frame with the highest <code>prop_X</code> ; NULL when no frame is positive.
<code>top_frame_prop</code>	DOUBLE PRECISION	Proportion of the top frame in the article.
<code>entropy_frames</code>	DOUBLE PRECISION	Shannon entropy (natural base) of the eight main-frame proportions; 0 when no main frame is positive.
<code>n_unique_messenger_subs</code>	SMALLINT	Number of distinct messenger sub-categories present in the article (0–9).
<code>n_unique_event_subs</code>	SMALLINT	Same for event sub-categories (0–8).
<code>n_unique_solution_subs</code>	SMALLINT	Same for solution sub-categories (0–2).

CCF_article_entities – article-level named-entity rollup

`CCF_article_entities` consolidates the sentence-level Named Entity Recognition output (the `ner_entities` JSON column in `CCF_processed_data`) into one row per article. The table is designed to answer the recurring analytical question *which entities does an article mention?* without forcing the analyst to parse the JSON for every sentence. It contains 266,271 rows and its structure is summarised in Table 8.

CCF_reliability_tiers – per-category quality lookup

`CCF_reliability_tiers` provides a single-row-per-category recommendation on how confidently each of the 65 annotation categories may be used in downstream analyses. The tier

Table 8: Columns of CCF_article_entities.

Column	Type	Definition
doc_id	BIGINT (PK)	Foreign key to CCF_full_data.
entities_per	JSONB	Deduplicated array of persons mentioned in the article.
entities_org	JSONB	Deduplicated array of organisations.
entities_loc	JSONB	Deduplicated array of locations.
n_unique_per, n_unique_org, n_unique_loc	INTEGER	Cardinalities of the three arrays above.
n_sentences_with_per, n_sentences_with_org, n_sentences_with_loc	INTEGER	Number of sentences in the article that mention at least one entity of the corresponding type.

assignment combines validation F_1 and blind-phase Cohen’s κ following the rules defined in Section ; the underlying metrics and flags are kept in the same row so that researchers can refine the tier rule for their own use case if they wish. Joining `CCF_processed_data` or `CCF_article_aggregates` to this table on the category code, and filtering on `tier_overall = 'A'`, restricts an analysis to the 27 research-grade categories. Columns are listed in Table 9.

CCF_sentence_embeddings – dense vector representations

`CCF_sentence_embeddings` enables semantic retrieval and similarity-based analyses across the whole corpus. It holds one row per sentence (and one extra row per article title, encoded with `sentence_id = 0`), for a total of 9,462,845 rows. Each row stores a 1024-dimensional BAAI/bge-m3 vector that is L2-normalised at encoding time, so cosine similarity reduces to a dot product and can be queried directly through the `pgvector <=>` operator. Storage uses the `halfvec(1024)` type (2 bytes per dimension, identical to the upstream `float16` representation), and an HNSW index on the cosine operator class delivers sub-second approximate top- k retrieval over the full table. Columns are summarised in Table 10.

The complete data dictionary, including every column type and the full set of join paths, is provided in Supplementary Table S12.

Data Overview

This section consolidates three reproducible summaries of the deposited CCF Database. Every figure and table is generated by the script `Results/Scripts/3_data_overview.py`, which pulls the underlying numbers directly from the deposited PostgreSQL tables: no manual transcription is involved, and re-running the script regenerates the artefacts byte-for-byte.

Article-level descriptive statistics. Table 11 reports the corpus size (article count, two-sentence analytical units, and per-article word statistics) computed on `CCF_full_data` and `CCF_processed_data`, and the mean per-article share of sentences positive for each annotation category, computed on the `prop_X` columns of `CCF_article_aggregates`. The

Table 11: Article-level descriptive statistics of the deposited CCF Database. Mean share is the average per-article proportion of sentences positive for the category; the top-frame column reports the percentage of articles for which the frame has the highest share among the eight frames. Computed from `CCF_full_data`, `CCF_processed_data`, and `CCF_article_aggregates`.

<i>Corpus: 266,271 articles, 9,198,958 two-sentence units; 732 words/article on average (SD = 531; median = 677).</i>				
Main frames			Primary categories & tone	
	Mean share (%)	Top frame (%)		Mean share (%)
Political frame	36.0	54.2	Messenger present	52.6
Economic frame	17.7	20.9	Event present	22.9
Scientific frame	8.0	9.4	Solution present	27.3
Environmental frame	3.5	4.4	Canadian context	46.4
Justice frame	3.7	1.4	Urgency to act	2.6
Cultural frame	3.6	7.2	Positive tone	9.5
Health frame	1.6	1.5	Negative tone	39.7
Security frame	0.5	0.4	Neutral tone	62.5

Figure 7: Mean per-article share of each main frame, year by year, 1980–2024. Each point is the mean of the corresponding `prop_X` column of `CCF_article_aggregates` across the articles published in that year, joined with `CCF_full_data` on `doc_id`. The pre-1988 portion of the curve is computed on fewer than fifty articles per year.

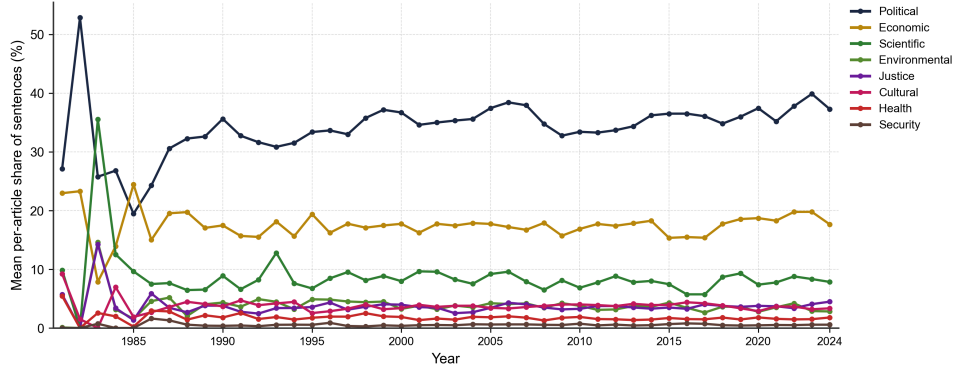


Table 12: Ten most frequently mentioned entities of each type in the deposited CCF Database. “Articles” is the number of articles in which the entity appears at least once. Computed from the JSONB arrays of `CCF_article_entities` (`entities_per`, `entities_org`, `entities_loc`).

#	Persons (PER)		Organisations (ORG)		Locations (LOC)	
	Entity	Articles	Entity	Articles	Entity	Articles
1	Stephen Harper	23,945	NDP	20,491	Canada	128,585
2	Justin Trudeau	23,542	United Nations	13,069	U.S.	58,999
3	Harper	19,922	UN	12,610	Ottawa	46,462
4	Trudeau	19,860	OTTAWA	9,535	Alberta	44,593
5	Barack Obama	11,870	Star	9,415	United States	36,582
6	Donald Trump	11,181	European Union	7,623	Ontario	34,157
7	Obama	8,616	Environment Canada	7,283	Toronto	33,717
8	Trump	8,523	House of Commons	7,103	U.S	22,499
9	George W. Bush	7,252	Senate	6,804	Quebec	21,967
10	Bush	6,822	National Post	6,222	Kyoto	20,614

Table 9: Columns of `CCF_reliability_tiers`.

Column	Type	Definition
<code>number</code>	SMALLINT (PK)	Supplementary Table S3 reference number (1–65).
<code>category</code>	TEXT	Readable category name.
<code>code</code>	TEXT (UNIQUE)	Column code as used in <code>CCF_processed_data</code> .
<code>tier_overall</code>	CHAR(1)	Tier on the combined corpus: “A”, “B”, or “C”.
<code>tier_en</code>	CHAR(1)	Tier on the English subset.
<code>tier_fr</code>	CHAR(1)	Tier on the French subset.
<code>paradox_kappa</code>	BOOLEAN	True if the category exhibits prevalence-induced reliability deflation ($F_1 \geq 0.70$ and $\kappa < 0.40$).
<code>exclusion_reason</code>	TEXT	“insufficient_training_data” for the two categories that could not be trained, NULL otherwise.
<code>f1_macro_all</code> , <code>f1_macro_en</code> , <code>f1_macro_fr</code>	DOUBLE PRECISION	Validation macro F_1 score on the gold standard, overall and per language.
<code>support_all</code> , <code>support_en</code> , <code>support_fr</code>	INTEGER	Number of positive (class 1) validation examples.
<code>cohens_kappa_blind_phase</code> , <code>gwet_acl_blind_phase</code> , <code>krippendorff_alpha_blind_phase</code>	DOUBLE PRECISION	Inter-coder agreement on the blind phase (sentences 601–1000).
<code>n_blind_phase</code>	INTEGER	Sentence count of the blind phase.
<code>prev_coder1_full</code> , <code>prev_coder2_full</code>	DOUBLE PRECISION	Prevalence (proportion of positives) per coder on the full 1,000-sentence sample.

Table 10: Columns of `CCF_sentence_embeddings`.

Column	Type	Definition
<code>doc_id</code>	BIGINT (PK, with <code>sentence_id</code>)	Foreign key to <code>CCF_full_data</code> .
<code>sentence_id</code>	INTEGER (PK)	Sentence position within the article. <code>sentence_id = 0</code> is reserved for the article title.
<code>embedding</code>	halfvec(1024)	L2-normalised BGE-M3 vector. Cosine distance is computed via the <code>pgvector <=></code> operator; an HNSW index on the cosine operator class is provisioned by the enrichment pipeline.

left block lists the eight frames together with the percentage of articles for which each frame is the `top_frame`; the right block lists the primary categories (messenger, event, solution, Canadian context, urgency) and the three tones.

Temporal coverage of the main frames. Figure 7 plots, for every calendar year between 1980 and 2024, the mean of each `prop_X` main-frame column of `CCF_article_aggregates`, grouped by the year extracted from `CCF_full_data.date`. No smoothing or regression is applied; each point is a yearly mean. The earliest years of the corpus carry

very few articles—fewer than fifty per year before 1988 and fewer than five hundred per year before 1990—so the pre-1988 portion of the curve is computed on small samples and should be read with that caveat.

Named-entity coverage. Table 12 reports the ten most frequently mentioned entities of each NER type. Entities are extracted from the JSONB arrays `entities_per`, `entities_org`, and `entities_loc` of `CCF_article_entities`, expanded with `jsonb_array_elements_text`, filtered to remove pure punctuation and one-character tokens, and counted in articles—an entity is therefore counted once per article regardless of how many sentences mention it. Variants such as *Trudeau* vs. *Justin Trudeau* are kept separate because the underlying NER pipeline emits them as distinct strings; analysts who need canonicalised entities can apply their own resolution step on top of these arrays.

Technical Validation

Technical validation of the CCF database and the trained models proceeded in three steps. First, a stratified validation set was constructed in line with established best practices for multi-label classification. Second, an expert annotator in climate change politics produced the reference annotations that define the gold standard used for all model evaluation. Third, an independent second annotator replicated the coding following a structured protocol comprising a training phase (sentences 1–600) and a blind coding phase (sentences 601–1000), enabling assessment of inter-coder reliability. The 1,000-sentence gold standard is strictly disjoint from the 4,000-sentence training/validation pool used in Phase 3. We explicitly excluded all sentence identifiers present in the training data before drawing the stratified sample, so that no test sentence was ever seen by a trained model at training time.

Construction of the stratified validation set

The validation set comprised 1,000 sentences (500 per language) extracted from the fully annotated database using a root-inverse probability weighting scheme combined with strict structural constraints. Each candidate sentence i received a sampling weight

$$w_i \propto \sum_{j \in C(i)} \frac{1}{\sqrt{f_j}}, \quad (1)$$

where $C(i)$ denotes the set of categories that are positive for sentence i and f_j denotes the corpus-wide prevalence of category j . The square root in the denominator deliberately moderated the influence of extremely rare categories, while still privileging them over common ones. Concretely, a sentence annotated with both *Economic Frame* (15.4% prevalence in the corpus) and *Loss of Indigenous Practices* (0.28% prevalence) received a weight of

$$\frac{1}{\sqrt{0.154}} + \frac{1}{\sqrt{0.0028}} = 2.55 + 18.90 = 21.45,$$

which was roughly eight times the weight of a sentence containing only *Economic Frame*, and ensured that under-represented categories nevertheless reached the validation set.

Two structural constraints were then applied on top of this weighted draw. First, a minimum threshold of 40 positive examples per category was imposed to guarantee sufficient statistical power for reliable performance estimation: without it, extremely rare categories such as *Disruption of military operations* (whose prevalence in the database is on the order of 10^{-5}) would have had too few examples to support a meaningful assessment of model accuracy. Second, a 35% maximum allocation per category was imposed to prevent any single prevalent category from monopolising the validation set; without this ceiling, *Actors/Messengers Detection* alone, present in 48.8% of database sentences, could plausibly have dominated the entire sample and left insufficient space for the rarer categories.

Expert gold-standard annotation and model performance

All 1,000 sentences in the validation set were manually annotated by an expert annotator specialising in climate change politics. These expert annotations defined the gold standard against which all model performance metrics reported in this section were computed.

The validation results demonstrated robust performance across all annotation dimensions, with an overall F1 macro score of 0.866 (0.869 for English, 0.864 for French). As detailed in Supplementary Table S7, primary detection categories achieved the highest performance, with *Canadian Context* (F1=0.982) and *Actors/Messengers* (F1=0.961) showing near-perfect classification. The frames maintained strong performance despite their semantic complexity, with *Scientific Frame* (F1=0.890) and *Environmental Frame* (F1=0.871) leading this category. The hierarchical classification strategy proved particularly effective, as evidenced by consistently higher recall scores for detection categories (mean recall=0.891) that triggered conditional sub-category evaluation. Notably, the rarest category in the validation set, *Disruption of military operations*, reached perfect precision on the French subset where two positive examples were available; the broader *Post-disaster military assistance* category, by contrast, had too few positives to support a reliable evaluation in either language and was flagged as exploratory in the reliability tiers (Supplementary Table S10).

To assess temporal stability, stratified validation was conducted across five temporal periods (1980s–2020s); the analysis revealed minimal performance drift ($\Delta F1 < 1.1\%$) despite

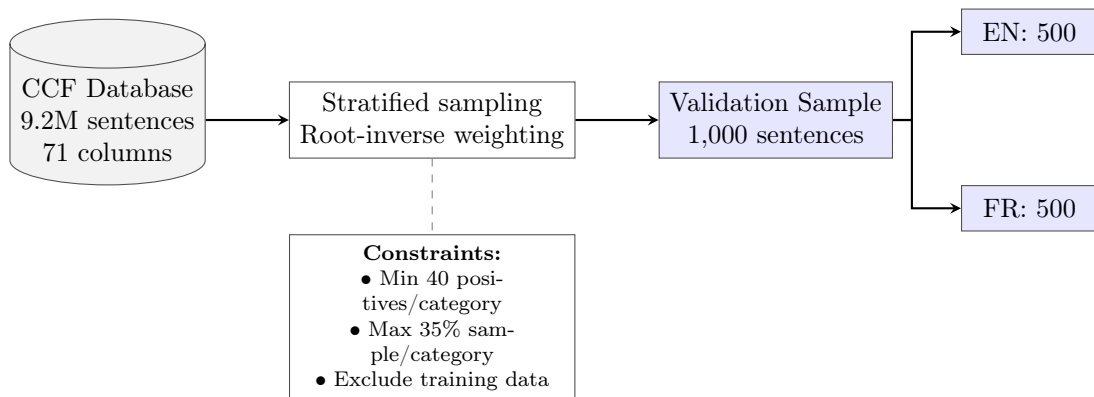


Figure 8: Stratified sampling procedure for final validation. Root-inverse probability weighting ensures balanced representation across all 65 annotation categories.

evolving media discourse patterns (Figure 9). This temporal consistency validated the models’ generalisation capacity across the full 47-year corpus span.

Table 13: Overall validation performance metrics

Language	F1 Macro	F1 Micro	F1 Weighted
English (EN)	0.869	0.909	0.911
French (FR)	0.864	0.902	0.905
Combined (ALL)	0.866	0.905	0.908

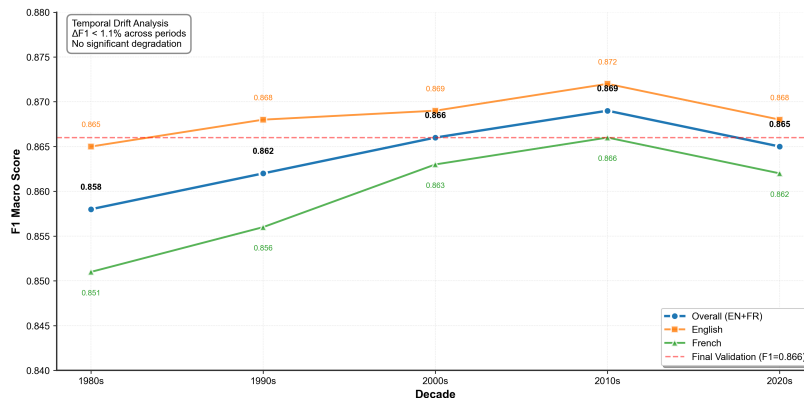


Figure 9: Temporal validation showing stable F1 macro scores across five time periods. The minimal variation ($\Delta F1 < 1.1\%$) confirms robust model performance without temporal degradation.

Inter-coder assessment of the gold standard

Inter-coder reliability assessment was completed with a second independent annotator evaluating the same 1,000 randomly selected sentences from the validation set. The protocol followed a structured approach: during the first 600 sentences, the expert annotator provided training to the second coder, explaining category definitions and answering questions without directly intervening in the coding process. This training phase allowed the second coder to progressively learn the annotation framework while maintaining independence in coding decisions. After sentence 600, the annotation process became completely blind, with no communication between coders. The resulting comparison between the second annotator’s labels and the expert gold-standard annotations provided an explicit evaluation of the reliability and reproducibility of the gold standard.

Given that the authors’ main interest lay in the level of reliability once the codebook had been learned, the metrics for the blind coding phase (sentences 601–1000) are reported as the primary benchmark. Table 14 summarises the agreement between the second annotator and the expert gold standard across all 65 annotation categories in this phase. These values indicated moderate agreement on Cohen’s κ and substantial agreement on Gwet’s AC1, while Krippendorff’s α exceeded the commonly used 0.667 threshold for acceptable reliability. In

other words, the gold-standard annotations produced by the expert could be reproduced to a reasonably high degree by an independent trained annotator, even in a setting with 65 multi-label categories and often dense and multi-topic climate discourse.

However, several limitations should be kept in mind. The moderate κ values reflected the inherent difficulty of climate discourse annotation at the sentence level, where many sentences contain subtle framings and overlapping categories. Chance-corrected indices such as Cohen’s κ are also known to be sensitive to category imbalance, which is pronounced in this setting. The substantially higher Gwet’s AC1 suggested that part of the apparent disagreement in κ was driven by these prevalence-related paradoxes rather than by systematic confusion between coders. A further limitation concerns the training data itself: as discussed in Phase 3, all 4,000 training sentences were coded by a single expert, with the second coder reserved for the independent gold-standard validation. This design is consistent with the standard recommendation of stabilising the codebook through one expert before validating on a held-out sample [38], but it nevertheless means that the model behaviour reflects, at training time, the interpretive choices of a single coder. The per-category reliability tiers (Supplementary Table S10) and the blind-phase agreement metrics (Supplementary Table S9) are designed to make these choices auditable; future releases of the CCF will extend the multi-coder protocol to a stratified subsample of the training data. A second limitation concerns the three pre-trained NER models (Supplementary Table S6), whose precision, recall, and F1 figures are reported from the original-model documentation rather than from a CCF-specific gold standard. A systematic in-domain NER evaluation against a manually coded sample is planned for a subsequent release of the database; users are accordingly advised to treat the named-entity rollup as a high-recall extraction layer rather than as a benchmarked classifier.

Table 14: Inter-coder reliability metrics for the blind coding phase (sentences 601–1000)

	Cohen’s κ	Gwet’s AC1	Krippendorff’s α
Blind coding phase (601–1000, $n = 400$)	0.596	0.894	0.698

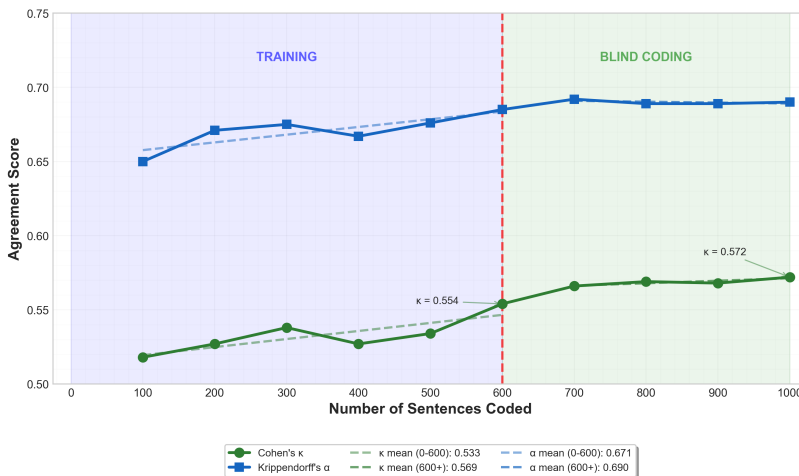


Figure 10: Inter-coder reliability progression across 1,000 annotated sentences, computed in successive blocks of 100 sentences.

To better understand how reliability evolved as the second annotator internalised the codebook, agreement metrics in successive blocks of 100 sentences were computed. This analysis showed a pattern consistent with a learning curve for a complex codebook: both Cohen’s κ and Krippendorff’s α increased over time, with higher values in the later blocks than in the early ones. Figure 10 illustrates this progression across the 1,000 sentences, with a continuing improvement after the transition to blind coding at sentence 600. The upward trajectory of both indices across the 100-sentence blocks indicated that, once training was complete, the annotation framework could be applied consistently. This reinforces the view that the expert-defined gold standard is reproducible by trained coders.

Validation of the database enrichment

Three integrity checks were run at the end of the enrichment pipeline to confirm that the derived tables remained exactly synchronised with the underlying sentence-level annotations.

First, the article-level aggregates were cross-validated against the source sentence table: for each of the 65 `prop_X` columns, the SQL identity `prop_X = SUM(COALESCE(X, 0)) / COUNT(*)` was verified to hold within floating-point tolerance (10^{-9}) for every one of the 266,271 articles, and the `top_frame` column was confirmed to equal the column code with the highest `prop_X` in every non-empty row. Second, the per-article entity rollup was checked by re-extracting the union of all distinct entities from the `ner_entities` JSON column for a stratified random sample of 1,000 articles and comparing the result element-wise against the JSONB arrays stored in `CCF_article_entities`; all 1,000 articles matched exactly. Third, the BGE-M3 embeddings were verified to have an L2 norm of 1.0 ± 10^{-3} on a random sample of 10,000 rows—the expected behaviour of vectors produced with `normalize_embeddings = True`—and an HNSW round-trip query (retrieve the top-1 nearest neighbour of an arbitrary vector and confirm that it is the vector itself) returned the expected sentence with cosine distance below 10^{-3} in every one of 100 spot-checks.

Authors Contribution Statement

Antoine Lemor designed and implemented the complete data architecture, including data extraction, cleaning, parsing, and database construction. Developed all machine learning models and methodological approaches. Wrote all code and conducted quantitative analyses. Drafted the methodology and results sections.

Alizée Pillod developed the theoretical and conceptual foundations underlying the annotation framework. Designed the complete annotation scheme and performed all manual annotation. Drafted the theoretical background and literature review sections.

Matthew Taylor contributed to conceptual development and analysis design. Verified statistical analyses. Reviewed and edited the complete manuscript.

Richard Nadeau contributed to the conceptual development of the project.

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